AI

Generated date: 2026-04-27
Team size: 4
Skill level: Beginner
Timeline: 3 days
Difficulty score: 6/10
Completion confidence: 68%

Project Flow (ASCII)

```
Project Idea
   |
Planning -> Development -> Testing -> Polish & Demo
   |
Final Deliverables
   |
Milestones Achieved
```

Executive Summary

This roadmap outlines a AI/GenAI project plan for AI with an estimated difficulty of 6/10 and completion confidence of 68%. The plan is structured for steady delivery, incremental validation, and final demo readiness.

Tools & Tech Stack

- Python
- LangChain
- Streamlit
- OpenAI API
- HuggingFace
- Pandas
- Jupyter

Day-wise Plan

DAY 1 - Execution

Task 1: Define scope and acceptance checks for Day 1

What: Map the day goals for AI into 4-5 concrete acceptance checks.

How: Create notes/day_1_scope.md and list measurable checks aligned with the current phase.

Output: A clear day scope document with acceptance checklist ready for execution.

Time: 30 mins

Task 2: Implement the primary execution component

What: Build the highest-priority feature for Day 1 tied to the project milestone.

How: Code the component using the selected stack, run a targeted smoke test, and capture logs/screenshots.

Output: Working feature implementation with test evidence and reproducible run steps.

Time: 90 mins

Task 3: Document progress and blockers for Day 1

What: Summarize completed work, pending items, and blockers found during execution.

How: Update notes/day_1_report.md with bullets, error traces, and next-day handoff notes.

Output: A review-ready daily report that can be used in demo/viva discussion.

Time: 25 mins

Task 4: Implement core execution code path in `src/day_1_feature.py`

What: Write or extend the main logic for AI in a dedicated module for Day 1.

How: Create `src/day_1_feature.py`, add `build_day_1_feature()` using Python, and run local test command.

Output: Module compiles and produces expected output for one realistic AI scenario.

Time: 1h 15 mins

Task 5: Test critical flow with CLI/API checks for Day 1

What: Verify the key feature path with positive and negative test cases.

How: Run `python -m pytest` (or equivalent) and capture response/trace details for failed and passing cases.

Output: Test evidence file with pass/fail status and corrected issues.

Time: 50 mins

Day duration estimate: 5 hours

Key output: Execution deliverable checkpoint 1

DAY 2 - Execution

Task 1: Define scope and acceptance checks for Day 2

What: Map the day goals for AI into 4-5 concrete acceptance checks.

How: Create notes/day_2_scope.md and list measurable checks aligned with the current phase.

Output: A clear day scope document with acceptance checklist ready for execution.

Time: 30 mins

Task 2: Implement the primary execution component (variation 2)

What: Build the highest-priority feature for Day 2 tied to the project milestone.

How: Code the component using the selected stack, run a targeted smoke test, and capture logs/screenshots.

Output: Working feature implementation with test evidence and reproducible run steps.

Time: 90 mins

Task 3: Document progress and blockers for Day 2

What: Summarize completed work, pending items, and blockers found during execution.

How: Update notes/day_2_report.md with bullets, error traces, and next-day handoff notes.

Output: A review-ready daily report that can be used in demo/viva discussion.

Time: 25 mins

Task 4: Implement core execution code path in `src/day_2_feature.py`

What: Write or extend the main logic for AI in a dedicated module for Day 2.

How: Create `src/day_2_feature.py`, add `build_day_2_feature()` using Python, and run local test command.

Output: Module compiles and produces expected output for one realistic AI scenario.

Time: 1h 15 mins

Task 5: Test critical flow with CLI/API checks for Day 2

What: Verify the key feature path with positive and negative test cases.

How: Run `python -m pytest` (or equivalent) and capture response/trace details for failed and passing cases.

Output: Test evidence file with pass/fail status and corrected issues.

Time: 50 mins

Day duration estimate: 5 hours

Key output: Execution deliverable checkpoint 2

DAY 3 - Execution

Task 1: Define scope and acceptance checks for Day 3

What: Map the day goals for AI into 4-5 concrete acceptance checks.

How: Create notes/day_3_scope.md and list measurable checks aligned with the current phase.

Output: A clear day scope document with acceptance checklist ready for execution.

Time: 30 mins

Task 2: Implement the primary execution component (variation 3)

What: Build the highest-priority feature for Day 3 tied to the project milestone.

How: Code the component using the selected stack, run a targeted smoke test, and capture logs/screenshots.

Output: Working feature implementation with test evidence and reproducible run steps.

Time: 90 mins

Task 3: Document progress and blockers for Day 3

What: Summarize completed work, pending items, and blockers found during execution.

How: Update notes/day_3_report.md with bullets, error traces, and next-day handoff notes.

Output: A review-ready daily report that can be used in demo/viva discussion.

Time: 25 mins

Task 4: Implement core execution code path in `src/day_3_feature.py`

What: Write or extend the main logic for AI in a dedicated module for Day 3.

How: Create `src/day_3_feature.py`, add `build_day_3_feature()` using Python, and run local test command.

Output: Module compiles and produces expected output for one realistic AI scenario.

Time: 1h 15 mins

Task 5: Test critical flow with CLI/API checks for Day 3

What: Verify the key feature path with positive and negative test cases.

How: Run `python -m pytest` (or equivalent) and capture response/trace details for failed and passing cases.

Output: Test evidence file with pass/fail status and corrected issues.

Time: 50 mins

Day duration estimate: 5 hours

Key output: Execution deliverable checkpoint 3

MVP Features

- Text input/output
- Basic memory

Optional Features

- Voice input
- Multi-language
- File upload

Risk Register

Risk	Severity	Mitigation
API rate limiting during peak use	High	Immediate mitigation owner, add fallback path, a
Prompt injection attacks	High	Immediate mitigation owner, add fallback path, a
Hallucinated responses	High	Immediate mitigation owner, add fallback path, a
Model output inconsistency	Medium	Track in sprint board and review in daily standu
Embedding drift over time	Medium	Track in sprint board and review in daily standu
Context window overflow	Medium	Track in sprint board and review in daily standu
Sensitive data leakage in prompts	High	Immediate mitigation owner, add fallback path, a
High inference latency	Medium	Track in sprint board and review in daily standu
Token cost spikes	Medium	Track in sprint board and review in daily standu
Low quality retrieval sources	High	Immediate mitigation owner, add fallback path, a
Evaluation metrics not aligned to users	Medium	Track in sprint board and review in daily standu
API rate limits	Medium	Track in sprint board and review in daily standu

Milestones Timeline

1. Basic chat working
2. Memory added
3. UI polished
4. Deployed

Viva Preparation

Focus on explaining architecture decisions, trade-offs, and risk handling for AI.

Q: What is RAG?

A: Prepare a concise 3-4 line technical explanation with one example.

Q: How does memory work in chatbots?

A: Prepare a concise 3-4 line technical explanation with one example.

Report Summary

This roadmap outlines a AI/GenAI project plan for AI with an estimated difficulty of 6/10 and completion confidence of 68%. The plan is structured for steady delivery, incremental validation, and final demo readiness.